

Computing Laplace–Beltrami Eigenbasis from a Neural SDF

1 Problem Setup

Assume that we have trained a neural signed distance function (SDF)

$$F_\theta : \mathbb{R}^3 \rightarrow \mathbb{R},$$

whose zero level set defines the target surface

$$\Gamma_\theta = \{x \in \mathbb{R}^3 \mid F_\theta(x) = 0\}.$$

We assume that Γ_θ is a regular surface, i.e.,

$$\|\nabla F_\theta(x)\| > 0 \quad \text{for } x \in \Gamma_\theta.$$

Our goal is to compute the Laplace–Beltrami eigenbasis on Γ_θ :

$$-\Delta_{\Gamma_\theta} \psi_k = \lambda_k \psi_k, \quad k = 0, 1, \dots, K-1,$$

with the $L^2(\Gamma_\theta)$ orthonormality constraint

$$\langle \psi_i, \psi_j \rangle_{L^2(\Gamma_\theta)} = \int_{\Gamma_\theta} \psi_i(x) \psi_j(x) dA = \delta_{ij}.$$

The first eigenfunction ψ_0 is constant and corresponds to $\lambda_0 = 0$.

2 Surface Geometry from the SDF

For a point $x \in \Gamma_\theta$, the surface normal is obtained from the SDF gradient:

$$n_\theta(x) = \frac{\nabla F_\theta(x)}{\|\nabla F_\theta(x)\|},$$

and the tangent-plane projection matrix is

$$P_\theta(x) = I - n_\theta(x) n_\theta(x)^\top.$$

We represent each eigenfunction by a scalar neural network $\phi_\eta : \mathbb{R}^3 \rightarrow \mathbb{R}$. Its restriction to the surface defines a scalar field

$$\psi_\eta = \phi_\eta|_{\Gamma_\theta}.$$

Since ϕ_η is a differentiable extension of the surface function to the ambient space $\mathbb{R}^3$, the surface gradient is obtained by projecting the Euclidean gradient onto the tangent plane:

$$\nabla_{\Gamma_\theta} \psi_\eta(x) = P_\theta(x) \nabla \phi_\eta(x) = \nabla \phi_\eta(x) - (\nabla \phi_\eta(x) \cdot n_\theta(x)) n_\theta(x). \quad (1)$$

We now compare this with the Spherical Neural Surface (SNS) formulation. In SNS, the surface is represented by an *explicit* map $S_\theta : S^2 \rightarrow \Gamma$, and tangent gradients are transformed using the inverse-transpose Jacobian $(J(S_\theta)^\top)^{-1}$. Because SNS is an explicit parametrization, it cannot be applied directly to a surface that is only available as the zero level set of an implicit field; this is precisely the gap addressed here. In the SDF setting no spherical parametrization is required: the surface geometry is encoded implicitly by F_θ , and tangent gradients are computed directly via the projection in Equation (1).

3 Rayleigh Quotient Formulation

For a scalar function ψ defined on Γ_θ , the Rayleigh quotient is

$$Q_{\Gamma_\theta}(\psi) = \frac{\int_{\Gamma_\theta} \|\nabla_{\Gamma_\theta} \psi(x)\|^2 dA}{\int_{\Gamma_\theta} |\psi(x)|^2 dA}. \quad (2)$$

There is no factor of $\frac{1}{2}$: on a closed manifold, Green's first identity gives

$$\int_{\Gamma_\theta} \psi_k (-\Delta_{\Gamma_\theta} \psi_k) dA = \int_{\Gamma_\theta} \|\nabla_{\Gamma_\theta} \psi_k\|^2 dA,$$

so that $\lambda_k = Q_{\Gamma_\theta}(\psi_k)$ when $\|\psi_k\|_{L^2(\Gamma_\theta)} = 1$.

The low-frequency eigenfunctions of the Laplace–Beltrami operator can be obtained by the variational (trace-minimization) problem

$$\psi_0, \dots, \psi_{K-1} = \arg \min_{\psi_0^*, \dots, \psi_{K-1}^*} \sum_{k=0}^{K-1} Q_{\Gamma_\theta}(\psi_k^*), \quad \text{s.t.} \quad \langle \psi_i, \psi_j \rangle_{L^2(\Gamma_\theta)} = 0, \quad i \neq j. \quad (3)$$

The eigenvalue of mode k is recovered as $\lambda_k \approx Q_{\Gamma_\theta}(\psi_k)$ after optimization. Note that Q_{Γ_θ} is invariant to the scale of ψ , so an explicit normalization constraint $\|\psi\|_{L^2} = 1$ is *not* needed to recover the eigenfunction; if used, such a term serves only to improve conditioning and does not change the minimizer.

4 Surface Integration: Coarea Formula and Monte Carlo

4.1 The coarea formula

For an analytic or meshed surface, the surface measure dA is available directly. For an implicit surface $\Gamma_\theta = \{F_\theta = 0\}$ no parametrization exists, and we instead obtain surface integrals from *volume* integrals through the coarea formula. For a thin shell $S_\epsilon = \{x : |F_\theta(x)| < \epsilon\}$,

$$\int_{S_\epsilon} h(x) dV = \int_{-\epsilon}^{\epsilon} \left(\int_{\{F_\theta=t\}} \frac{h(x)}{\|\nabla F_\theta(x)\|} dA_t \right) dt.$$

Choosing $h = g \|\nabla F_\theta\|$ and letting $\epsilon \rightarrow 0$,

$$\int_{\Gamma_\theta} g(x) dA \approx \frac{1}{2\epsilon} \int_{S_\epsilon} g(x) \|\nabla F_\theta(x)\| dV. \quad (4)$$

4.2 Monte Carlo estimator

Sampling $\{x_i\}$ uniformly in a bounding box B and keeping those with $|F_\theta(x_i)| < \epsilon$, the accepted shell points are projected onto Γ_θ by a Newton step

$$\hat{x}_i = x_i - \frac{F_\theta(x_i)}{\|\nabla F_\theta(x_i)\|^2} \nabla F_\theta(x_i), \quad (5)$$

after which the gradient is re-evaluated at $\hat{x}_i$ to obtain the unit normal n_i and the area-measure weight

$$w_i = \|\nabla F_\theta(\hat{x}_i)\|. \quad (6)$$

By Equation (4), every surface integral becomes a weighted sum $\int_{\Gamma_\theta} g dA \propto \sum_i w_i g(\hat{x}_i)$, and the constant prefactor cancels in the Rayleigh quotient. The weighted Rayleigh quotient and the weighted inner product are therefore

$$\widehat{Q}_{\Gamma_\theta}^w(\psi_\eta) = \frac{\sum_i w_i \|P_\theta(\hat{x}_i) \nabla \phi_\eta(\hat{x}_i)\|^2}{\sum_i w_i |\phi_\eta(\hat{x}_i)|^2}, \quad \widehat{\langle \psi_\eta, \psi_j \rangle}^w = \frac{\sum_i w_i \phi_\eta(\hat{x}_i) \psi_j(\hat{x}_i)}{\sum_i w_i}. \quad (7)$$

5 Algorithm

We train all K neural fields $\phi_{\eta_1}, \dots, \phi_{\eta_K}$ jointly; the constant mode ψ_0 is fixed. Each eigenfunction is modeled as the composition of a neural field $\bar{\phi}_i$ and a projection operator $\mathcal{P}_i$,

$$\phi_i = \mathcal{P}_i \circ \bar{\phi}_i. \quad (8)$$

Unlike the explicit orthogonalize-then-normalize operator used for volumetric spectra in Shape Space Spectral, $\mathcal{P}_i$ here consists of a single step, because Q_{Γ_θ} is scale-invariant (Section 3) and no unit-norm projection is needed:

$$\mathcal{P}_i : \quad \bar{\phi}_i \xrightarrow{\text{weighted Gram-Schmidt}} \phi_i^\perp.$$

Weighted Gram-Schmidt projection. Any two eigenfunctions of Δ_{Γ_θ} corresponding to distinct eigenvalues must be $L^2(\Gamma_\theta)$ -orthogonal,

$$\langle \psi_i, \psi_j \rangle_{L^2(\Gamma_\theta)} = 0, \quad i \neq j,$$

a constraint we enforce by Gram-Schmidt orthogonalization within each training iteration. Given a candidate network $\bar{\phi}_k$ that is not yet orthogonal to the k modes $\psi_0, \dots, \psi_{k-1}$ already fixed earlier in the iteration, we subtract from $\bar{\phi}_k$ its component already spanned by these dominating modes. Concretely, we seek the linear combination of $\psi_0, \dots, \psi_{k-1}$ that best approximates $\bar{\phi}_k$ on Γ_θ ,

$$q^\star = \arg \min_{q \in \mathbb{R}^k} \left\| \sum_{j=0}^{k-1} q_j \psi_j - \bar{\phi}_k \right\|_{L^2(\Gamma_\theta)}^2, \quad (9)$$

and subtract the resulting projection:

$$\psi_k^\perp = \bar{\phi}_k|_{\Gamma_\theta} - \sum_{j=0}^{k-1} q_j^\star \psi_j. \quad (10)$$

By construction $\psi_k^\perp$ is orthogonal to every $\psi_0, \dots, \psi_{k-1}$ under the $L^2(\Gamma_\theta)$ inner product.

We estimate the domain integral defining Equation (9) by the same weighted stochastic cubature used for the Rayleigh quotient (Section 4). Let $\{\hat{x}_i\}_{i=1}^N$ be the batch points and $W = \text{diag}(w_1, \dots, w_N)$ the area-measure weights from Equation (6); let $U \in \mathbb{R}^{N \times k}$ collect the previous modes at these points, $U_{ij} = \psi_j(\hat{x}_i)$, and $r \in \mathbb{R}^N$ the candidate values $r_i = \bar{\phi}_k(\hat{x}_i)$. The discretized problem (9) is the weighted linear least squares

$$q^\star = \arg \min_{q \in \mathbb{R}^k} \|W^{1/2}(Uq - r)\|_2^2, \quad (11)$$

whose normal equations and residual are

$$(U^\top W U) q^\star = U^\top W r, \quad \psi_k^\perp(\hat{x}_i) = r_i - (Uq^\star)_i. \quad (12)$$

The weighting is essential: replacing it with the unweighted $U^\top U q = U^\top r$ would estimate orthogonality with respect to the sampling density $\rho \propto 1/\|\nabla F_\theta\|$ rather than the true area measure on Γ_θ (Remark 1), so the resulting $\psi_k^\perp$ would not be $L^2(\Gamma_\theta)$ -orthogonal to the previous modes.

The tangential gradient is projected with the *same* coefficients $q^\star$,

$$\nabla_{\Gamma_\theta} \psi_k^\perp(\hat{x}_i) = P_i \nabla \phi_{\eta_k}(\hat{x}_i) - \sum_{j=0}^{k-1} q_j^\star \nabla_{\Gamma_\theta} \psi_j(\hat{x}_i), \quad (13)$$

which keeps the automatic-differentiation path through the projection consistent: $\psi_k^\perp$ and its surface gradient are obtained by the same linear operation on the network outputs, so back-propagation through Equation (12) flows correctly into η_k .

Algorithm 1: Joint optimization of K neural Laplace–Beltrami eigenfunctions on Γ_θ

Input : Trained SDF F_θ ; number of modes K

Output: Neural eigenfunctions $\phi_{\eta_1}, \dots, \phi_{\eta_{K-1}}$ and eigenvalues $\lambda_1, \dots, \lambda_{K-1}$

Build weighted surface pool $\{(\hat{x}_i, n_i, w_i)\}$ from F_θ // Eqs. (5), (6); Sec. 4

Initialize K Fourier-feature MLPs $\bar{\phi}_1, \dots, \bar{\phi}_K : \mathbb{R}^3 \rightarrow \mathbb{R}$

$epoch \leftarrow 0$

repeat

 Sample batch $(\hat{x}, n, w)$ from the pool; rescale so $\sum_i w_i = N$

 Rank the K networks by $\hat{Q}_{\Gamma_\theta}^w$ (ascending) // causal sorting, Eq. (7)

$\psi_0 \leftarrow \mathbf{1}$ // constant eigenfunction; $U \leftarrow [\psi_0]$

for k in causal-sorted order **do**

 Evaluate $\bar{\phi}_k(\hat{x})$ and, by autodiff, $\nabla \bar{\phi}_k(\hat{x})$

 Form $P_i = I - n_i n_i^\top$ and surface gradient $P_i \nabla \bar{\phi}_k$ // Eq. (1)

 Solve $(U^\top W U) q = U^\top W r$ with $r = \bar{\phi}_k(\hat{x})$, set $\psi_k^\perp = r - Uq$ // weighted

 Gram–Schmidt, Eq. (12)

 Project the surface gradient with the same q // Eq. (13)

 Compute $\mathcal{L}(\eta_k)$ and backpropagate // Eq. (14)

 Concatenate $\psi_k^\perp$ into U and detach

$epoch \leftarrow epoch + 1$

until $epoch = \text{MaxEpoch}$

Estimate $\lambda_k \approx \hat{Q}_{\Gamma_\theta}^w(\psi_k^\perp)$ for each k

Per-mode loss. Each network is updated by minimizing the weighted Rayleigh quotient of its orthogonalized output, plus a soft orthogonality penalty:

$$\mathcal{L}(\eta_k) = \hat{Q}_{\Gamma_\theta}^w(\psi_k^\perp) + \lambda_{\text{ortho}} \frac{\sum_i w_i ((Uq)_i)^2}{\sum_i w_i (\psi_k^\perp(\hat{x}_i))^2}. \quad (14)$$

After training, the eigenvalue of mode k is estimated as $\lambda_k \approx \hat{Q}_{\Gamma_\theta}^w(\psi_k^\perp)$.

Causal sorting. At every iteration the K networks are re-ranked by their current weighted Rayleigh quotient $\hat{Q}_{\Gamma_\theta}^w$ (Equation (7)) in ascending order. We took the idea from Shape Space Spectral